

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 5

Exercise 5.1 (Logistic map).

Consider the discrete (non-invertible) dynamical system obtained by iteration of the logistic map

$$f : [0, 1] \rightarrow [0, 1], \quad f(x) = rx(1 - x)$$

for some $r \in [0, 4]$.

- a) Find the fixed points and check if they are hyperbolic.
- b) Show that there exists a 2-periodic orbit for $r = 4$.

Solution.

- a) We solve $rx(1 - x) = x$ to find $x = 0$ and $x = \frac{r-1}{r}$. The latter only lies in $[0, 1]$ if $r \geq 1$ and they coincide only for $r = 1$. Hence, we have only one fixed point at 0 for $r \leq 1$ and a second one at $\frac{r-1}{r}$ for $r > 1$. We further compute $f'(x) = r - 2rx$, so at the fixed points we have $f'(0) = r$ and $f'(\frac{r-1}{r}) = 2 - r$. Consequently $x = 0$ is hyperbolic as long as $r \neq 1$ while $x = \frac{r-1}{r}$ is hyperbolic for $r \in (1, 4] \setminus \{3\}$ (for $r=1$ this fixed point coincides with 0).
- b) *Remark:* The existence of a 2-periodic orbit can not only be shown for $r = 4$, but every $r > 3$. The calculations however might become quite cumbersome.

2-periodic points satisfy $f(f(x)) = x$, i.e., they are fixed points of the second iterate. Hence, we need to solve $f(f(x)) = r \cdot rx(1 - x)(1 - rx(1 - x)) = x$ or, equivalently, $r \cdot rx(1 - x)(1 - rx(1 - x)) - x = 0$, which is a quartic equation. However, since we already know that every fixed point of f also satisfies $f(f(x)) = x$ we can reduce this problem to a quadratic one by factoring out x and $x - \frac{r-1}{r}$. For $r = 4$ one obtains, that the remaining two solutions are $x = \frac{1}{8}(5 \pm \sqrt{5}) \in [0, 1]$. In fact, $f(\frac{1}{8}(5 \pm \sqrt{5})) = \frac{1}{8}(5 \mp \sqrt{5})$.

To only obtain the existence of these 2-periodic points one can also use that $f'(0)$ and $f'(\frac{r-1}{r})$ are larger than 1. In fact, defining

$$g : [0, 1] \rightarrow [0, 1], \quad g(x) = f(f(x)) - x,$$

we obtain that $g(0) = g(\frac{r-1}{r}) = 0$. Then,

$$g'(0) = f'(0)^2 - 1 > 0 \quad \text{and} \quad g'\left(\frac{r-1}{r}\right) = f'\left(\frac{r-1}{r}\right)^2 - 1 > 0.$$

This implies that g is positive for small values larger than $x = 0$ and negative for close smaller values of $\frac{r-1}{r}$. The intermediate value theorem states that there exists a solution of $f(f(x)) = x$ in $(0, \frac{r-1}{r})$. Since 2-period points need to have a fixed point in-between, we conclude that the other point that solves $g(x) = 0$ lies in the interval $(\frac{r-1}{r}, 1)$.

□

Exercise 5.2 (No oscillations in 1d flows).

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function and consider the ODE $\dot{x} = f(x)$ with $x \in \mathbb{R}$. Prove that there does not exist a periodic orbit

- a) by basic tools from an introductory calculus lecture, (e.g. mean-value theorem),
- b) by considering $\int_0^T f(x(t)) \frac{dx(t)}{dt} dt$,
- c) by exploiting the existence of a potential $V: \mathbb{R} \rightarrow \mathbb{R}$ such that $-\nabla V(x) = f(x)$.

Solution.

- a) Assume there was a periodic orbit $x(t)$ with period $T > 0$. Let $t_1 = \operatorname{argmin}_{0 \leq t \leq T} x(t)$ and $t_2 = \operatorname{argmax}_{t_1 \leq t \leq t_1+T} x(t)$. Moreover denote $x_1 = x(t_1)$ and $x_2 = x(t_2)$. We have $t_2 < t_1 + T$ since $x(t_1 + T) = x(t_1) = x_1$. By the mean-value theorem, there exists a $\xi_1 \in (t_1, t_2)$ such that $\dot{x}(\xi_1) = (x_2 - x_1)/(t_2 - t_1) > 0$. Moreover, by the intermediate theorem, there is at least one $\xi_2 \in (t_2, t_1 + T)$ with $x(\xi_2) = x(\xi_1)$. Since $x^{-1}(\{x_1\})$ is a closed set, there exists a minimum of all possible ξ_2 that satisfy this property. Let ξ_2 be the minimal such ξ_2 . Then, $x(t) > x(\xi_2)$ for all $t \in (t_2, \xi_2)$. Consequently, $f(x(\xi_2)) = \dot{x}(\xi_2) \leq 0$. However, we also have $f(x(\xi_2)) = f(x(\xi_1)) = \dot{x}(\xi_1) > 0$, which is a contradiction.
- b) Assume the existence of a periodic orbit with period $T > 0$. Then, we observe that $\int_0^T f(x(t)) \frac{dx(t)}{dt} dt = \int_0^T (f(x(t)))^2 dt \geq 0$ with equality only if $f(x(t)) = 0$ for all $t \in [0, T]$, as $f(x(t))$ is continuous. However, then every point would be an equilibrium and there would not be a periodic orbit. Thus equality cannot occur. On the other hand $\int_0^T f(x(t)) \frac{dx(t)}{dt} dt = \int_{x(0)}^{x(T)} f(x) dx = \int_{x(0)}^{x(0)} f(x) dx = 0$, by the rule of substitution. This is a contradiction, so there cannot be a periodic orbit.
- c) Let $V: \mathbb{R} \rightarrow \mathbb{R}$ with $V(x) = -\int_0^x f(\hat{x}) d\hat{x}$. Then, $-\nabla V(x) = f(x)$. Consequently, $\frac{d}{dt} V(x(t)) = \nabla V(x(t)) \cdot \dot{x}(t) = -f(x(t))^2 \leq 0$ and V decreases along trajectories. It strictly decreases if $\dot{x}(t) \neq 0$, which has to be the case on a periodic orbit. This contradicts the fact that $V(x(t)) = V(x(t+T))$.

□

Exercise 5.3 (Poincaré map).

Consider the planar ODE system

$$\begin{aligned} x' &= \mu x - y - x(x^2 + y^2), \\ y' &= x + \mu y - y(x^2 + y^2), \end{aligned}$$

where $\mu > 0$ is a parameter.

Find a periodic orbit and determine its stability using a Poincaré map.

Hint: Use polar coordinates.

Solution.

Transforming the system into polar coordinates $(x, y) = (r \cos(\theta), r \sin(\theta))$ we obtain

$$\begin{aligned} r' &= \mu r - r^3, \\ \theta' &= 1. \end{aligned} \tag{5.1}$$

Observe that the angular velocity is constant and independent of the radius. Furthermore, the radial equation is independent of the angle and has a equilibrium point at $r = \sqrt{\mu}$ (and at $r = 0$). As a consequence we have a periodic orbit given by the circle around the origin with radius $\sqrt{\mu}$. Since $\theta' = 1$ this periodic orbit has period 2π .

For a Poincaré map we can consider for example a section S around $(\sqrt{\mu}, 0)$ perpendicular to the periodic orbit. Since $\theta' = 1$ everywhere, the return time for any point of this section is 2π . The Poincaré map is thus given by the time- 2π -map of the continuous flow of (5.1), i.e., $P((r, 0)) = \phi_{2\pi}(r, 0)$ for any point $(r, 0) \in S$. Along one revolution the radial component is always governed by the equation

$$r' = \mu r - r^3 = f(r). \quad (5.2)$$

We find the linearization at $r = \sqrt{\mu}$ to be $Df(\sqrt{\mu}) = \mu - 3\sqrt{\mu}^2 = -2\mu < 0$, as we assumed that $\mu > 0$. This entails that

$$\phi_t(r, 0) = \left(\sqrt{\mu} + e^{-2\mu t} (r - \sqrt{\mu}) + o(r - \sqrt{\mu}), 0 \right),$$

for r close to $\sqrt{\mu}$, and that

$$P(r, 0) = \left(\sqrt{\mu} + e^{-4\mu\pi} (r - \sqrt{\mu}) + o(r - \sqrt{\mu}), 0 \right).$$

Consequently, the linearized Poincaré map on $r = \sqrt{\mu}$ has unique eigenvalue $-1 < e^{-4\mu\pi} < 1$ and $\sqrt{\mu}$ is stable in S under the Poincaré map. In conclusion, the periodic orbit is stable. $\square$

Exercise 5.4 (Poincaré Return Map Multipliers).

Let $N \in \mathbb{N}$ with $N \geq 2$ and $\mathbb{T}^N$ be the N -dimensional torus defined by $\mathbb{T}^N := (\mathbb{R}/2\pi\mathbb{Z})^N$. Consider the ODE

$$\phi'_k = \omega + \frac{K}{N} \sum_{l=1}^N \left(\frac{1 + \delta \sin(5\phi_l)}{1 + \delta \sin(5\phi_k)} \sin(\phi_l - \phi_k + \alpha) - \sin(\alpha) \right), \quad \text{for } k = 1, \dots, N,$$

where $\phi = (\phi_1, \dots, \phi_N) \in \mathbb{T}^N$. Moreover, $\omega > 0$, $0 \leq \delta < 1$ and $K \in \mathbb{R}$ are parameters. The dynamical system induced by this ODE has a periodic orbit $\Gamma(t): \mathbb{R} \rightarrow \mathbb{T}^N$ with $\Gamma_k(t) = \omega t$ (independent of k) for $k = 1, \dots, N$ and period $T = 2\pi/\omega$.

- Let $\delta = 0$. Calculate the multipliers of a Poincaré return map and infer the stability of Γ .
- Now, let δ be arbitrary. Calculate the multipliers of a Poincaré return map and infer the stability of Γ .

Hint: The eigenvalues of

$$\exp \left(\frac{a}{N} (\mathbb{1} - N\mathbb{I}) \right),$$

where $a \in \mathbb{R}$, $\mathbb{1} = \{1\}^{N \times N}$ is the matrix full of ones, $\mathbb{I} \in \mathbb{R}^{N \times N}$ is the identity matrix and $\exp$ is the matrix exponential, are given by e^{-a} with multiplicity $N - 1$ and 1 with multiplicity 1.

Solution.

We will only present the solution of b), as this includes a).

Let us define a function $f: \mathbb{T}^N \rightarrow \mathbb{R}^N$ as

$$f_k(\phi) = \omega + \frac{K}{N} \sum_{l=1}^N \left(\frac{1 + \delta \sin(5\phi_l)}{1 + \delta \sin(5\phi_k)} \sin(\phi_l - \phi_k + \alpha) - \sin(\alpha) \right), \quad \text{for } k = 1, \dots, N.$$

Then, the ODE can be written as $\phi' = f(\phi)$ with $\phi = (\phi_1, \dots, \phi_N) \in \mathbb{T}^N$. We proceed as in the lecture and consider the variational equation

$$u'(t) = A(t)u(t), \quad A(t) = Df(\Gamma(t)).$$

Calculating the derivative Df yields

$$A(t) = g(t) \frac{K}{N} (\mathbb{I} - N\mathbb{I}),$$

with

$$g(t) = \frac{1}{1 + \delta \sin(5\omega t)} \left(5\delta \cos(5\omega t) \sin(\alpha) + (1 + \delta \sin(5\omega t)) \cos(\alpha) \right).$$

Since $A(t)$ commutes with $A(s)$ for all t, s , we can use Exercise 1.2 b) to calculate a solution of the variational equation. In particular, the monodromy matrix $M(T)$, that satisfies $u(T) = M(T)u(0)$, is given by

$$\begin{aligned} M(T) &= \exp \left(\int_0^T A(s) ds \right) \\ &= \exp \left(\int_0^T g(s) \frac{K}{N} (\mathbb{I} - N\mathbb{I}) ds \right) \\ &= \exp \left(\int_0^T g(s) ds \frac{K}{N} (\mathbb{I} - N\mathbb{I}) \right) \end{aligned}$$

Integrating $g(t)$ over one period yields

$$\begin{aligned} \int_0^T g(s) ds &= [\sin(\alpha) \log(1 + \delta \sin(5\omega s)) + s \cos(\alpha)]_{s=0}^{s=T} \\ &= T \cos(\alpha). \end{aligned}$$

We proceed to calculate $M(T)$:

$$M(T) = \exp \left(T \cos(\alpha) \frac{K}{N} (\mathbb{I} - N\mathbb{I}) \right)$$

Using the hint, it follows that the multipliers of $M(T)$ are 1 with multiplicity 1 and $e^{-TK \cos(\alpha)}$ with multiplicity $N - 1$. By Theorem 5.6, the multipliers of a Poincaré map are $e^{-TK \cos(\alpha)}$ with multiplicity $N - 1$. Consequently, the periodic orbit Γ is stable if $K \cos(\alpha) > 0$ and unstable if $K \cos(\alpha) < 0$. $\square$